Small Area Estimation with the Fay-Herriot Model

ICAR and Logit Transformation

Fall 2026
07.11.2026

University of Colorado Denver

Brady Lamson
brady.lamson@ucdenver.edu

Contents

1. Introduction 1
1.1. Background 1
2. The Data 1
2.1. Data Sources 1
2.1.1. 2020 Decennial Census of Population and Housing 1
2.1.2. American Community Survey 1
2.1.3. NASS Crop Data Layer 2
2.2. Data Processing 2
2.2.1. American Community Survey 2
2.2.1.1. Logit Transformation of the Response 3
2.2.1.2. Logit Transformation of the Standard Error 3
2.2.2. 2020 Decennial Census of Population and Housing 4
2.2.3. NASS Crop Data Layer 5
2.3. Data Analysis 5
2.3.1. Iowa Spatial Structure 5
2.3.2. 2020 Decennial Census of Population and Housing 6
2.3.3. American Community Survey 7
2.3.4. Tests of Spatial Autocorrelation for Housing Vacancy Rates 10
2.3.5. NASS Crop Data Layer 10
3. Methods 13
4. Results 13
4.1. Summary Results 13
4.1.1. Mean Squared Error and Mean Absolute Error 13
4.1.2. Mean Standard Deviation and Mean Interval Width 14
4.1.3. Interval Coverage 14
4.1.4. Continuous Ranked Probability Score (CRPS) 15
4.2. Crop Data Layer Models 15
4.2.1. Coefficients 15
4.2.2. Bayes Factor 15
4.3. Model Estimate comparisons 16
4.3.1. Residuals and Standard Deviations 16
4.3.2. Interval Coverage Outcomes 18
5. Conclusion 19
Bibliography 19

1. Introduction

1.1. Background

Much of the data that statisticians rely on for their work is taken for granted.

2. The Data

2.1. Data Sources

2.1.1. 2020 Decennial Census of Population and Housing

The first source of data for this project comes from the 2020 Decennial Census. Mandated by the Constitution of the United States, the census is carried out every 10 years by the U.S. Census Bureau and "determines the number of people living in the United States" [1]. Though the primary purpose of the census is to distribute seats in the U.S. House of Representatives [1], it is also the official source of high level demographic and housing data for the entire country. As such, it is used as the source of truth in this project. Every single model will be compared to the values provided by the 2020 Decennial Census to evaluate the quality of predictions. It is important to note that, though treated as the source of truth, that the values provided by the U.S. census are not perfect. An analysis performed by the U.S. Government Accountability Office in 2024 found 14 states exhibiting statistically significant net coverage errors in the 2020 census, though Iowa is absent from this list [2]. Accounting for this uncertainty is outside the scope of this project, so all values provided by the census are treated as the true values.

2.1.2. American Community Survey

The second, and primary source of data for this project comes from the American Community Survey (ACS). The ACS is an ongoing survey performed by the U.S. Census Bureau about the population of the United States of America [3]. The Census Bureau is primarily known for the U.S. Decennial Census, but the lesser known ACS serves as a crucial companion piece to it. Instead of every 10 years, the ACS is sent out to a sample of addresses every single month of every single year [4]. The ACS in this aspect fills a very important role, as it is able to provide current information between decennial census collections, and is also able to cover a more robust set of topics than the census. The tradeoff is that the ACS comes from a sample of households, so every result is met with a certain degree of uncertainty. However, unlike traditional surveys, individuals selected for the ACS are legally obligated to respond to it [5]. The consequence of this is that the uncertainty in the ACS results can be treated as a known quantity, which allows for mathematical techniques that are not normally possible with traditional surveys that are optional to respond to.

The data from the ACS comes in two primary forms, 1-year and 5-year estimates. The 1-year estimates are less precise but more current, encompassing 12 months of collected data but only providing information on areas with a population size of at least $65,000$ [6]. The 5-year estimates are based on 60 years of collected data, and so are more precise but less

current. The 5-year estimates also provide information on all areas surveyed, so this even includes extremely sparsely populated county or census tract level data [6].

This project will be using the 2020 5-year estimates, encompassing survey data from 2016-2020 for each county in Iowa. The variables used are `B25002_001` and `B25002_003` which correspond to the number of total housing units and vacant housing units respectively. The ratio of the two is used to calculate a vacancy rate for each county in Iowa which functions as the response for this project.

2.1.3. NASS Crop Data Layer

The third source of data for this project is known as the Crop Data Layer, or CDL. According to the USGS, the CDL, created by the USDA National Agricultural Statistics Service (NASS), is "an annual raster, geo-referenced, crop-specific land cover data layer produced using satellite imagery and extensive agricultural ground reference data" [7]. The USDA website carries direct downloads to every annual data layer going back to 2008. The year used for this analysis is the same as the decennial census, 2020. This dataset contains per-pixel cropland classifications at a 30-meter resolution for the entire country. The classifications were done using a machine learning model trained primarily on the Farm Service Agency (FSA) Common Land Unit (CLU) data, though some states incorporate additional auxiliary data from other agencies and institutions [7]. This data within the region of Iowa is used to create the spatial covariate for the Fay-Herriot model. Incorporation of pixel classification uncertainty in the CDL is outside the scope of this project, so all classifications are treated as the true classifications.

2.2. Data Processing

2.2.1. American Community Survey

ACS data was acquired for this project through the Census Bureau API, indirectly interfaced with via the `tidycensus` R package on CRAN. Data was pulled from the 2020 5-year ACS estimates for the `B25002_001` and `B25002_003` variables, which as previously mentioned correspond to the total housing units and vacant housing units per county. The resulting dataframe has a grain of one row per variable per county, so 2 rows per county. Each row contains the estimate for that variable, its corresponding Margin of Error (MoE) and the spatial geometry of that county. This dataframe was pivoted to a grain of one row per county to facilitate creation of the response variable, vacancy rates. For each county, the vacant unit count was divided by the total housing unit count to get the vacancy rate of that county.

$$\text{Vacancy Rate} = \left(\frac{\text{Vacant Housing Units}}{\text{Total Housing Units}}\right)$$

To get the standard deviation of this new estimate, two steps are necessary. First, a conversion from MoE to standard error is required for each estimate. Then, the standard errors of both estimates must be combined to form the standard error of the proportion. The Census Bureau

provides guidelines on both steps, and the same document will be referred to for both [8]. Conversion from MoE to SE is done by simplying dividing the MoE by 1.645.

$$\text{Standard Error} = \frac{\text{Margin of Error}}{1.645}$$

As for the standard error of the proportion response, the following setup is used. First, let $P = \frac{A}{B}$, where A is a subset of B. A will represent the vacant housing units, B will be represent all housing units. Then, the standard error of a counties housing vacancy rate can be represented as

$$\text{SE}(P) = \frac{1}{B}\sqrt{\text{SE}(A)^2 - P^2 \times \text{SE}(B)^2}$$

2.2.1.1. Logit Transformation of the Response

As the Fay-Herriot model relies on linear regression as a key component, a proportion response can behave poorly. The domain of the output of a linear reression is the entire real line, whereas a proportion is bounded between 0 and 1. In a more traditional scenario this would lead to a logistic regression model, though usage of the Fay-Herriot model requires alternative methods to overcome this limitation. Logit transformations have been used in this scenario before, such as in research performed by Cucu Sumarni et al. that found a "logit transformation within the SAE framework improved the precision of proportion estimates" [9]. This takes a response bounded between 0 and 1 and maps it to the entire real line. Thus, our transformed response can be represented as

$$Q = \ln\left(\frac{P}{1-P}\right)$$

2.2.1.2. Logit Transformation of the Standard Error

Yet again, additional work is required for the conversion of the standard error. For this, the delta method was utilized. For reference,

Definition 2.2.1.2.1 (Delta Method)**.** Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \to N(0, \sigma^2)$ in distribution. For a given function g and a specific value of θ, suppose that $g'(\theta)$ exists and is nonzero. Then

$$\sqrt{n}[g(Y_n) - g(\theta)] \to N\left(0, \sigma^2[g'(\theta)]^2\right)$$

in distribution.

To apply this, let

$$g(P) = \ln\left(\frac{P}{1-P}\right)$$

Then,

$$g'(P) = \frac{1}{P - P^2}$$

We can then substitute this into the definition to get the updated variance, σ_Q^2.

$$\sigma_Q^2 = \sigma^2[g'(P)]^2)$$
$$= \sigma^2\left(\frac{1}{P - P^2}\right)^2$$
$$= \frac{\sigma^2}{(P - P^2)^2}$$
$$= \frac{\sigma^2}{P^2(1 - P)^2}$$

From this, our transformed standard error is

$$\sigma_Q = \frac{\sigma}{P(1 - P)}$$

The final dataframe contains a variety of other columns for diagnostic and exploratory purposes, but the dataframe of modeling components looks like so:

County	Vacancy Rate	SE	Vacancy Rate (Logit)	SE (Logit)	Geometry
Polk	0.064	0.003	−2.69	0.05	-
Grundy	0.076	0.011	−2.50	0.154	-
-	-	-	-	-	-

Table 1: Snippet of ACS Dataframe used for modeling. Untransformed columns and geometry included for the sake of reference and demonstration.

2.2.2. 2020 Decennial Census of Population and Housing

Ingestion and processing of the 2020 decennial census data was handled in a similar manner to the ACS data. Data was acquired through the `tidycensus` package, used to access the 2020 Census Demographic and Housing Characteristics File (DHC). The variables pulled were `H3_001N` and `H3_003N`, which correspond to total housing units and total vacant units repsectively. The grain of the data is the same as the original ACS data, one row per county per variable. The final table had one row per county, and a new column representing the housing vacancy rate was created in percent units. These values were not passed through a LOGIT transformation as they are intended for comparison against the predictions after they have been transformed back into the original proportion scale.

To ensure the accuracy of census housing vacancy rates, output was compared to the official housing vacancy numbers provided by the Census Bureau in their 2020 Iowa breakdown [10]. All of the values matched their values presented on that sites dashboard. Below is a snippet of the data.

County	Total Housing Units	Vacant Housing Units	Vacancy Rate
Polk	210184	13293	0.06
Grundy	5465	396	0.07
-	-	-	-

Table 2: Snippet of the Census DataFrame containing the true values for model evaluation. Non-proportion columns retained for the sake of reference and demonstration.

2.2.3. NASS Crop Data Layer

The crop data layer requires careful processing to be usable in this project. At a 30×30 meter resolution, the 2020 crop data layer raster image has a rectangular dimension of 96523×153811 pixels, or 14.8 billion pixels total. About 6.2 billion (42%) pixels are empty background pixels and 8.7 billion encompassing actual categorizations. Naive calculations performed on this object are prone to overloading the memory of the machine running them. The `terra` package was used for loading and processing this file. For filtering down to only Iowa's portion of the CDL, the `tigris` package was utilized to download Iowa's TIGER shapefiles from the U.S. Census. The coordinate reference systems (CRS) of the two objects were aligned to NAD83 Conus Albers (ESPG:5070). Cropping and masking is then performed on the CDL using the Iowa shapefile object. Cropping filters the CDL down to the rectangular region containing Iowa, and masking changes all pixels outside of the state borders to NA.

This new object, the Iowa CDL, contains a rectangular dimension of 11671×17794, or approximately 207 million pixels total. That is 1.3% the size of the original CDL. Of those, about 46 million (22%) are empty background, and appoximately 162 million (78%) pixels are categorized. From here, the goal is to get a table containing the frequency and proportion breakdown of every county in the state. To accomplish this goal, a similar process was used as before. County shapefiles were acquired using `tigris` and, using an iterative approach over the counties that leverages cropping and masking, the frequencies of every CDL classification were captured alongside the total number of pixels within the county's borders. A dataframe was created from this that contains one row per county, with the percent of the pixels in the county that were classified as either corn or soybeans. The proportion from this functions as the spatial covariate for the model and can be represented as:

$$\text{Proportion Corn or Soy}_i = \frac{\text{\# of Corn Pixels}_i + \text{\# of Soybean Pixels}_i}{\text{\# of Pixels Total}_i}$$

With i representing the index of a given county.

Below is a snippet of the resulting dataframe. Values are saved as percentages and converted to proportions at the modeling stage.

County	Soybeans (%)	Corn (%)	Corn or Soy (%)
Polk	17.8	22.4	40.2
Grundy	36.0	52.5	88.5
-	-	-	-

Table 3: Snippet of the CDL DataFrame containing the spatial covariate of the model. The individual soy and corn columns are kept for reference.

2.3. Data Analysis

2.3.1. Iowa Spatial Structure

Iowa was chosen for this project for a few reasons. The first is that it has some very populated counties and very sparsely populated counties. This allows for a good demonstration of the

Fay-Herriot model as it can effectively pool from informative counties. This will be discussed in more detail in later sections.

Second is the layout of its counties. Iowa has 99 counties total and they are arranged in a very neat grid as shown in the figure below. On average, each county has 5.9 neighbors with a minimum of 3 and a maximum of 8. There are no disjointed islands or bizarre geometries that would complicate spatial calculations as well. As such, this state shows useful properties for both Fay-Herriot modeling and spatial methods.

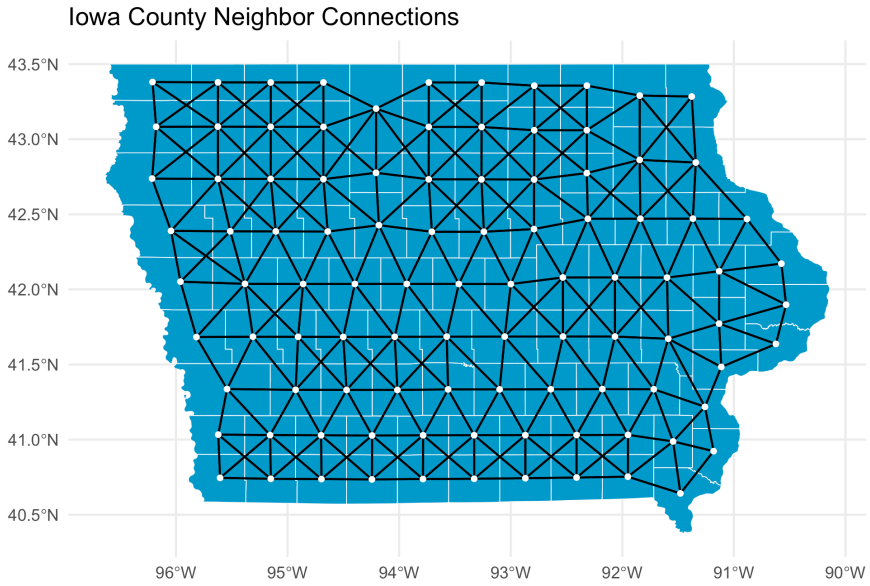


Figure 1: Plot showing adjacent county links across Iowa.

2.3.2. 2020 Decennial Census of Population and Housing

The counties in Iowa vary dramatically in terms of their population and housing. The number of total housing units can range from as little as 1,888 (Adams County) all the way up to 210,184 (Polk County). 29 of the 99 total counties have less than 5,000 total housing units.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	1888	274	0.05950
1st Quartile	4684	518	0.07892
Median	7037	701	0.10144
Mean	14271	1255	0.10842
3rd Quartile	10947	1131	0.11803
Maximum	210184	13293	0.41743

Table 4: Summary of the housing data from the 2020 decennial census. (N=99)

The same trend is seen with vacant units, as one would expect places with more houses also have more vacant houses. These variables show an almost perfect correlation ($r = 0.947$). This relationship does not hold however for vacancy rates which show a mild negative correlation with total housing units ($r = -0.237$). The highest vacancy rates appear on counties

with low total housing units. This is unsurprising as each vacant house contributes more to the vacancy rate than one would in larger counties.

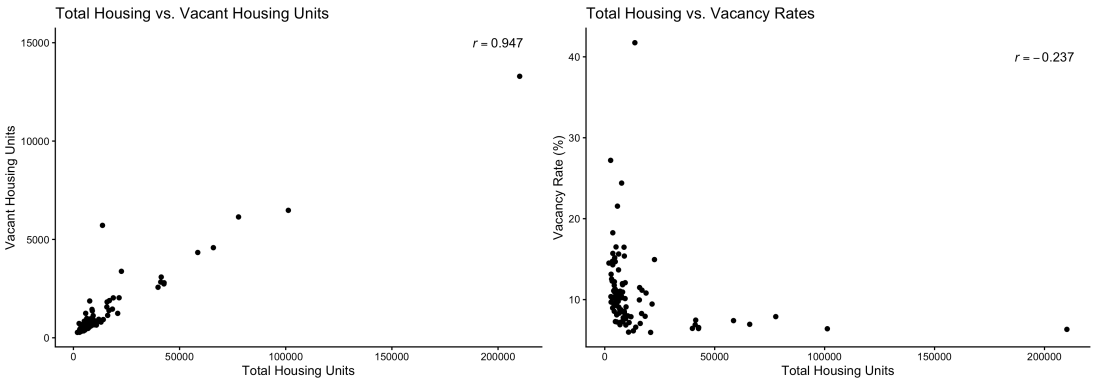


Figure 2: Plots showing the change of relationship when moving from pure vacant housing units to a vacancy rate.

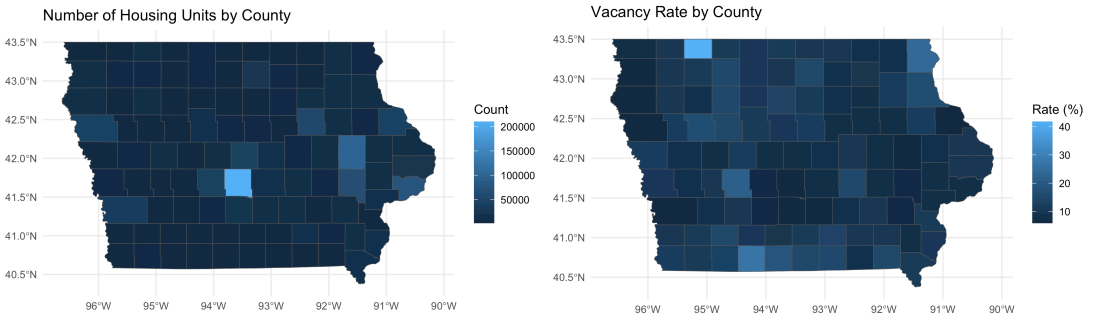


Figure 3: Spatial plots showing the distribution of housing units and vacancy rates by county across the state.

Examining these values at the spatial level, the overall population of housing units appears heavily concentrated in a small number of counties scattered throughout the state. In contract, vacancy rates show a far more mixed distribution of values. Visually it is unclear if there is evidence of spatial autocorrelation for the vacancy rates, but there does appear to be some grouping.

2.3.3. American Community Survey

Analysis begins with the ACS equivalent of Table 4, shown in Table 5. What is shown are near identical values, though there is a slight difference. The minimums never reach the same lows, and the maximums never reach the same highs. However, the values throughout the 1st and 3rd quartiles are consistently above those from the census.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	2011	296	0.0500
1st Quartile	4934	610.5	0.0900
Median	7199	776	0.0107
Mean	14220	1352.3	0.1210
3rd Quartile	11048	1370	0.1422
Maximum	204110	12985	0.3964

Table 5: Summary of the housing data estimates from the ACS. (N=99)

Examining the summary of the errors in Table 6 shows some extremely interesting behavior. The standard errors of the vacancy estimates are far higher than the total unit estimates. On average, the ratio of the vacant standard error to the total unit standard error is around 2.8, and goes all the way up to 7.1. This is before even accounting for the fact that the vacant unit estimates are far smaller, being a subset of the total. What this means is the vacancy unit estimates are far less precise than the total estimates are, and this lack of precision carries to the proportion as well. The relative standard errors for the vacancy estimates appear nearly identical to those of the vacancy rates.

Metric	Total SE	Vacant SE	Rate SE	Relative Total SE	Relative Vacant SE	Relative Rate SE
Minimum	14.59	43.77	0.003	0.000959	0.02784	0.02782
Q1	27.36	79.64	0.011	0.003855	0.10164	0.10164
Median	42.55	103.34	0.011	0.005198	0.12230	0.12230
Mean	50.85	129.06	0.014	0.005782	0.12231	0.12231
Q3	60.49	142.55	0.017	0.006975	0.14533	0.14532
Maximum	265.65	618.24	0.027	0.024768	0.25127	0.25127

Table 6: Summary of the estimate standard errors and relative standard errors from the ACS. The relative columns are calculated using (SE / Estimate) for each given estimate. Some adjustments to metric and column names made to preserve space.

Examining the relationships between the estimates, their corresponding standard errors and relative standard errors in Figure 4 shows us a few things. Firstly, it shows that the magnitude of standard errors tend to rise alongside the magnitude of the estimates themselves, and that this is consistent across all 3 types of estimates. Second, it shows that the opposite relationship arises when the standard errors are instead looked at relative to the size of the estimates themselves. In other words, larger estimates tend to be more precise. The counties with less housing units have larger errors in proportion to their estimates. This intutively makes sense as the samples themselves are smaller, there is less data to go off of for the estimates. It is worth noting that this does not mean that the larger estimates are more accurate, as no comparison has been made yet to the true values from the census.

The takeaway from these findings is that the smaller counties are going to be worth looking at closely in the model comparisons. In theory, these are the counties with the most

room for growth as they can pull from the higher precision in other counties with more housing units.

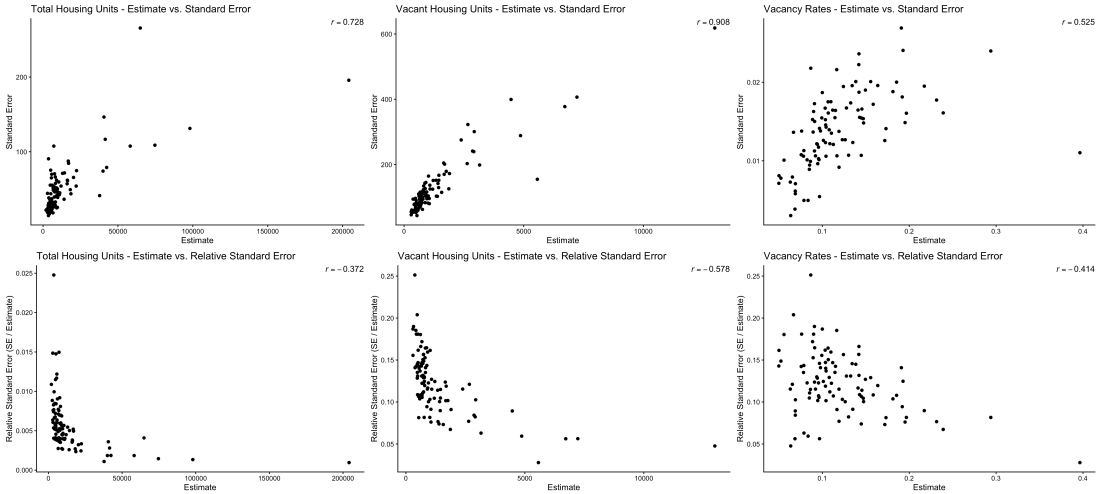


Figure 4: Plots showing the relationship estimates share with their standard errors and relative standard errors (SE / Estimate).

The distributions of the response and errors can be seen in Figure 5. The untransformed response is unimodal with a long right tail with a small bump in density around 0.4. The transformed response is very similar in distribution, but with a slightly less extreme right tail. Overall, it looks slightly more normally distributed than the untransformed response.

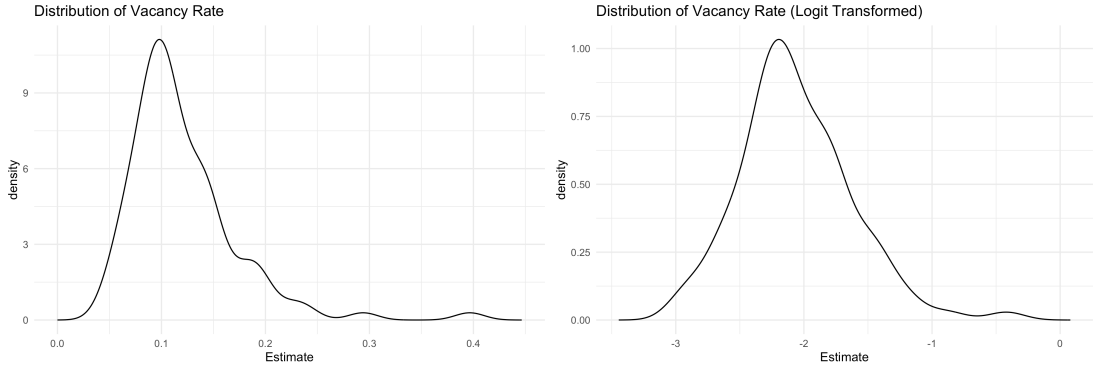


Figure 5: Distribution of the response before and after the logit transformation.

As for the errors shown in Figure 6, the shape stays mostly the same both before and after the response is transformed. The errors of the logit transformed response do show a couple additional bumps that aren't present in the original errors, though that may be more of an artifact of the plotting methodology than a meaningful change in distribution.

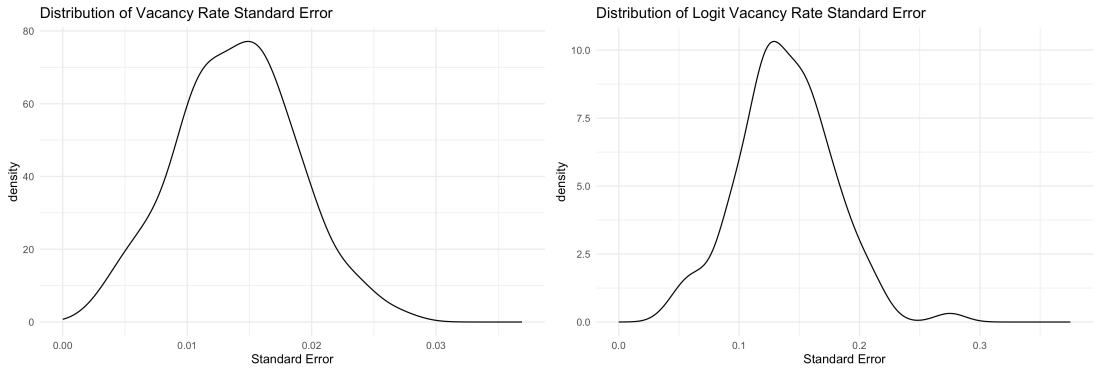


Figure 6: Distribution of the standard errors before and after the logit transformation of the response.

2.3.4. Tests of Spatial Autocorrelation for Housing Vacancy Rates

To assess the utility of an ICAR component in the Fay-Herriot model, Moran's I tests were performed on both the ACS response and the Census response separately to test for spatial autocorrelation. For this, the `moran.test` function was used from the `spdep` package on CRAN. For each dataset, two tests were performed under the constant risk hypothesis. One test uses binary neighbor encoding, another uses row standardization. The hypotheses for these tests are as follows:

$$H_0 : \rho = 0$$

$$H_a : \rho > 0$$

From the results in the figure below, both forms of encoding from both datasets reach the same conclusions at the $\alpha = 0.05$ level. There does seem to be evidence of positive autocorrelation of housing vacancy rates under the constant risk hypothesis.

Data	Matrix Encoding	Moran's I	Expected I	Variance	Z-Score	p-value
ACS	Binary	0.147	−0.01	0.0034	2.79	0.0026
ACS	Row Standardized	0.160	−0.01	0.0034	2.94	0.0016
Census	Binary	0.130	−0.01	0.0034	2.48	0.0066
Census	Row Standardized	0.142	−0.01	0.0034	2.62	0.0043

Table 7: Moran's I results from both the ACS and Census datasets. These tests performed under the normality assumption and constant risk hypothesis.

It is worth noting that these tests were primarily used as a diagnostic and a quick check of method feasibility. The risk from incorrectly rejecting the null hypothesis is low, and so additional tests were not performed. What these results show is that an ICAR component in the Fay-Herriot model does have some grounding, it isn't being applied to a response with no evidence of spatial autocorrelation.

2.3.5. NASS Crop Data Layer

To help motivate many of the decisions made in Section 2.2.2, analysis will begin with the crop data layer itself. Examining Table 8 shows that the makeup of land in Iowa is very different from that of the USA as a whole. Corn and Soybeans make up 38% and 27% of

the states land coverage respectively. Adding those up, approximately 65% of Iowas land coverage is corn or soybeans. Neither of these even reach 5% of the land coverage in the USA overall. The plots shown in Figure 7 visualize the similar spatial distributions and land coverage of corn and soybeans across the state. Their similarities and high prevelance, alongside their importance to Iowas overall agriculture gave the motivation for using them as a proxy for agricultural coverage per county.

USA		**Iowa**	
Crop	**Percentage**	**Crop**	**Percentage**
Shrubland	23.88	Corn	38.29
Grassland/Pasture	16.24	Soybeans	26.52
Evergreen Forest	12.58	Grassland/Pasture	13.73
Deciduous Forest	10.93	Deciduous Forest	8.43
Corn	4.85	Developed/Open Space	3.52
Woody Wetlands	4.59	Developed/Low Intensity	1.79
Soybeans	4.39	Herbaceous Wetlands	1.34

Table 8: Top crops by percentage of land classification for both the full USA CDL and Iowa subset. Top 7 rows.

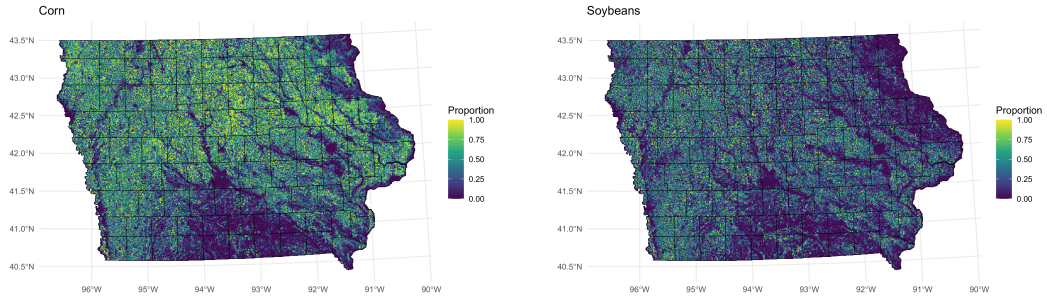


Figure 7: Plots showcasing the aggregated spatial distribution of corn and soybeans. Aggregation performed on each crop separately taking the mean of every 25x25 pixel area.

Amongst the various CDL classifications, there are ones that are referred to as "double crops". These are classifications with two different crops in them. There are 6 of these classifications total, examples including values like corn/soybeans and soybeans/oats. These were considered for inclusion in the covariate, though upon further inspection they were far too infrequent to justify. All 6 values total up to only around $4,000$ pixels total across all of Iowa. The max percentage for a double crop in a county was 0.042%. As such, these were ignored in all later work.

The top crops were examined for all counties to check for any other crops worth consideration. The top 3 crops for all 99 counties simplify down to only 6 cdl classifications. Those being corn, soybeans, developed/low intensity, developed/open space, grassland/pasture and deciduous forest. Nothing here appeared to be worth consideration for inclusion, so the decision was made to stick exclusively to corn and soybeans.

Narrowing down to the key crops at the county level, the overview in Table 9 shows that even the minimums show a large percentage of land coverage for the two key crops. At the most extreme almost 90% of land is classified as either corn or soybeans.

Statistic	Corn (%)	Soybeans (%)	Corn or Soybeans (%)
Minimum	8.58	6.53	19.95
1st Quartile	29.53	21.14	52.51
Median	41.32	27.47	69.36
Mean	37.73	26.42	64.15
3rd Quartile	47.73	32.22	77.84
Maximum	57.5	38.46	88.51

Table 9: Summary of the crop distribution across all Iowa counties.

Visualizing this information in Figure 8 shows some indication of spatial autocorrelation of land coverage. Many of the highest coverage counties are grouped together near the north west part of the state, and the lowest coverage counties are grouped in the south.

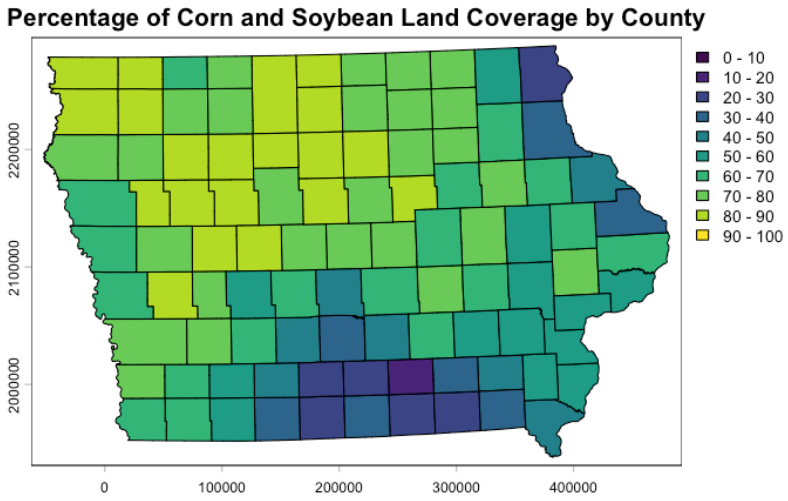


Figure 8: Plot of land coverage per county.

The plot in Figure 9 showcases a big problem with this choice of covariate. It appears to lack any meaningful relationship with the response. The points on the plot appear very randomly scattered, with no obvious trend to point to. This plot begs the question of why this is even a covariate for the model. This is something that will be explained in more detail in the limitations section, however to keep it brief this is the result of a change in topic partway through the project. The previous response had a far stronger relationship with the crop data layer. Due to the extensive amount of work done on this dataset, it is kept in this report primarily as a showcase of methodology.

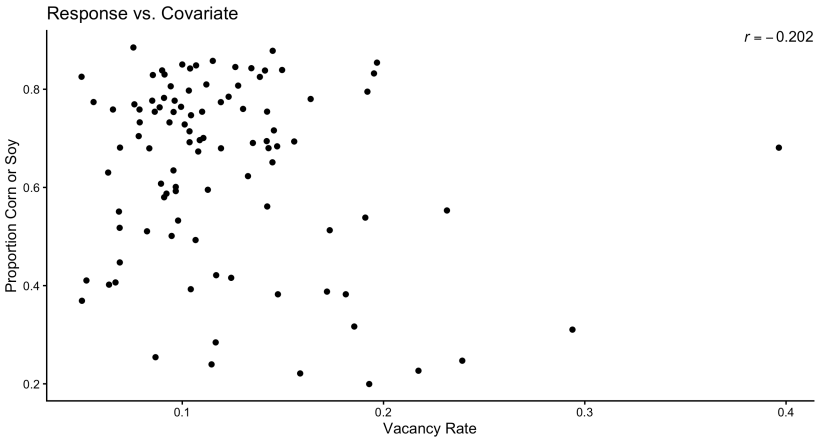


Figure 9: Scatterplot showing the lack of a relationship between vacancy rates and corn/soy land coverage.

3. Methods

4. Results

4.1. Summary Results

Table 10 contains all of the metrics used to gauge overall model performance. All results should be interpreted with respect to the ACS results, as those function as the baseline to beat. It is important to note that this section focuses on high level aggregated metrics. Further sections will focus more on performance at the individual estimate level.

Model	MSE	MAE	Mean SD	Interval Width	Interval Coverage	Mean CRPS
ACS	0.000503	0.01697	0.01395	0.05497	0.7374	—
Int (Freq)	0.000417	0.01586	—	—	—	—
Int (Bayes)	0.000436	0.01633	0.01329	0.05197	0.7576	0.011782
CDL	0.000438	0.01636	0.01331	0.05203	0.7576	0.011787
ICAR	0.000437	0.01644	0.01307	0.05113	0.7576	0.011786
Full Model	0.000433	0.01638	0.01306	0.05107	0.7677	0.011721

Table 10: Summary metrics for all models. "Int" refers to intercept only FH models. Int (Freq) is the frequentist model fit using `Emdi` whereas Int (Bayes) was fit using `brms`. Interval width represents the mean interval width for all counties.

4.1.1. Mean Squared Error and Mean Absolute Error

Starting with MSE and MAE, all FH models show improvement over the ACS estimates for both metrcs. The best performer here is actually the frequentist intercept only model, having the lowest value for both metrics. Though it is possible that this is genuinely the best model for MAE and MSE, it is worth considering that this could indicate incorrect estimation collection from the Bayesian models. In other words, that this is a symptom of user error. Regardless, moving on to the Bayesian models they all perform similarly. The full model

technically performs the best for MSE and the CDL model performs the best for MAE. The magnitude of these differences is miniscule however, and is not evidence on its own for any of the Bayesian models being the best on these metrics alone.

4.1.2. Mean Standard Deviation and Mean Interval Width

Moving onto standard deviation and interval width, some meaningful differences begin to take form. To get the baseline values for interval width, the logit transformed estimates and corresponding standard error were utilized to create a 95% interval on the original proportion scale. This was done by using those logit values to get the endpoints of the normal interval and then using the inverse logit function to transform them onto the proportion scale. Let $\hat{\theta}_i$ represent the logit transformed estimate for a county i and $\hat{\sigma}_i$ it's corresponding standard error.

$$\text{Lower}_i = \text{logit}^{-1}(\hat{\theta}_i + z_{0.025}\hat{\sigma}_i)$$

$$\text{Upper}_i = \text{logit}^{-1}(\hat{\theta}_i + z_{0.975}\hat{\sigma}_i)$$

From this, the baseline interval width for a given county is:

$$\boxed{W_i = \text{Upper}_i - \text{Lower}_i}$$

where $z_{0.025}$ and $z_{0.975}$ represent the 0.025 and 0.975 quantiles of the standard normal respectively.

All 4 models show improvement from that baseline on these 2 metrics, showcasing the increased precision expected from FH estimates. The two models that incorporate the ICAR component show further improvement on these metrics than the intercept only and CDL model. This shows some initial evidence that incorporating that spatial autocorrelation helps with estimate precision beyond the standard Fay-Herriot model. The frequentist model is unable to be compared on these metrics due the lack of uncertainty provided by its county estimates. This is the key upside of the Bayesian approach and provides an interesting tradeoff given the results shown so far. The Bayesian models are worth consideration even if the MSE and MAE performance is truly less than the frequentist approach as the uncertainty built into the estimates is extremely valuable for small area estimates such as these.

4.1.3. Interval Coverage

As for interval coverage, it represents the proportion of counties where the true value from the census is contained within the 95% intervals provided by the model estimates. The baseline value is 0.7374, or 73/99 counties where the ACS intervals captures the true value. All of the models show improvement here, going up to 75/99 counties for 4 of the models and 76/99 counties for the full model. So not only are the estimates more precise with smaller intervals, those intervals capture more of the counties true vacancy rates. There is a tradeoff here that will be examined in a later section. At a glance it appears more counties are being

captured that previously weren't with no downsides, though as will be shown later there are counties that the ACS captured that the Bayesian models lose.

4.1.4. Continuous Ranked Probability Score (CRPS)

For one final comparison of overall model performance, the Continuous Ranked Probability Score (CRPS) was captured for every prediction. The CRPS can be thought of as a generalisation of the mean absolute error (MAE) used for probabilistic outputs [11]. It's useful as it evaluates the accuracy of the overall distribution for each prediction instead of just the point estimate.

> The Continuous Ranked Probability Score (CRPS) is a generalisation of the Mean Absolute Error (MAE) tailored to probabilistic forecasting. Unlike the MAE, which compares point predictions to observations, the CRPS evaluates the accuracy of an entire predicted probaility distribution against the observed value. It does this by comparing the empirical cumulative distribution function (CDF) of the predicted values with the step-function CDF of the true value.
> Skforecast Documentation

As for the results of this score, all 4 bayesian models are nearly identical here. The full model pulls slightly ahead, but it is difficult to ascertain if the difference here is a meaningful one. It is worth noting that the intercept only model, which had the best MAE, loses out to the full model on CRPS.

4.2. Crop Data Layer Models

4.2.1. Coefficients

Before moving on to deeper model results, the Crop Data Layer Models must be investigated. As shown in Figure 9, there is a lack of an obvious relationship between the CDL covariate and housing vacancy rates. This raises the question on its ability to be an effective covariate for modeling. Table 11 shows the coefficients for both models featuring the CDL covariate. Examining the percentile columns for rows with the `Crop Proportion` variable provides additional evidence against this covariate. The 95% credible intervals for both the CDL and Full Model include 0, indicating that there is insufficient evidence of a non-zero effect.

Model	Variable	μ	σ	2.5%	97.5%
CDL	Crop Proportion	−0.385	0.244	−0.861	0.101
CDL	County Random Effects	0.427	0.034	0.366	0.501
CDL	Intercept	−2.054	0.045	−2.141	−1.967
Full Model	Crop Proportion	−0.602	0.426	−1.444	0.229
Full Model	ICAR Random Effects	0.888	0.072	0.758	1.039
Full Model	Intercept	−2.054	0.014	−2.082	−2.025

Table 11: Model Coefficients for models containing the Crop Data Layer Covariate.

4.2.2. Bayes Factor

TODO: DO THIS EXPLANATION BETTER DEAR GOD

Further investigation utilized the Bayes Factor of each model combination, shown in Table 12. The Bayes Factor is the ratio of the marginal likelihoods of the two selected models.

This comparison was done using the `bayes_factor` function from the `brms` package. For each combination of models, 5 bayes factors were calculated and then averaged together for the final result. This was on the recommendation of the package authors themselves, stating that this value can be unstable in some circumstances due to the bridge sampling it uses. Observed factors were stable across runs, though this methodology was preserved.

Model A	Model B	Bayes Factor (A/B)	Bayes Factor (B/A)
intercept	cdl	1.16730432	0.857
intercept	icar	0.00000025	3939449.516
intercept	full	0.00000022	4489470.694
cdl	icar	0.00000022	4533867.02
cdl	full	0.00000019	5207079.246
icar	full	0.87626698	1.141

Table 12: Bayes Factor for each model combination. Each row represents the mean of 5 separate bayes factor calculations.

Of relevance for this section are the rows where the CDL is added as a covariate where it was not there previously. Those rows are the first and last rows. These represent adding the CDL to the intercept only model, and adding the CDL to the ICAR model. The values, 1.167 for intercept vs. cdl and 1.141 for full vs. icar don't show a lot of evidence to indicate it's adding a lot of information.

The Bayes Factor in this project is being used primarily as a supplementary diagnostic instead of its typical role as an important hypothesis testing tool. There are many limitations to this methodology that make it unreliable for decision making.

Anyway, blech ugh god. Wont be adding in cdl stuff in later sections please thanks bye.

Probably cite like gelman (page 183) and others idk.

4.3. Model Estimate comparisons

4.3.1. Residuals and Standard Deviations

The plots in Figure 10 show how the residuals and standard deviations change across the ACS, Intercept Only and ICAR models. Visual inspection shows a reduction in standard deviation across both models, with a more varied reduction for the ICAR model. The overall trend is the same across both models. As the magnitude of the ACS standard deviation increases, the magnitude of the reduction does as well.

For the residuals, there is more of a tossup in results. Many of the counties show a reduction in the absolute residuals for both models. For the intercept only residuals, they tend to stick a bit more closely to the reference line compared to the ICAR model. Both Fay-Herriot models show consistent reductions in the residuals for the counties with larger residual magnitudes from the ACS. Lower magnitude counties are less consistent however, with updated residuals moving in both directions.

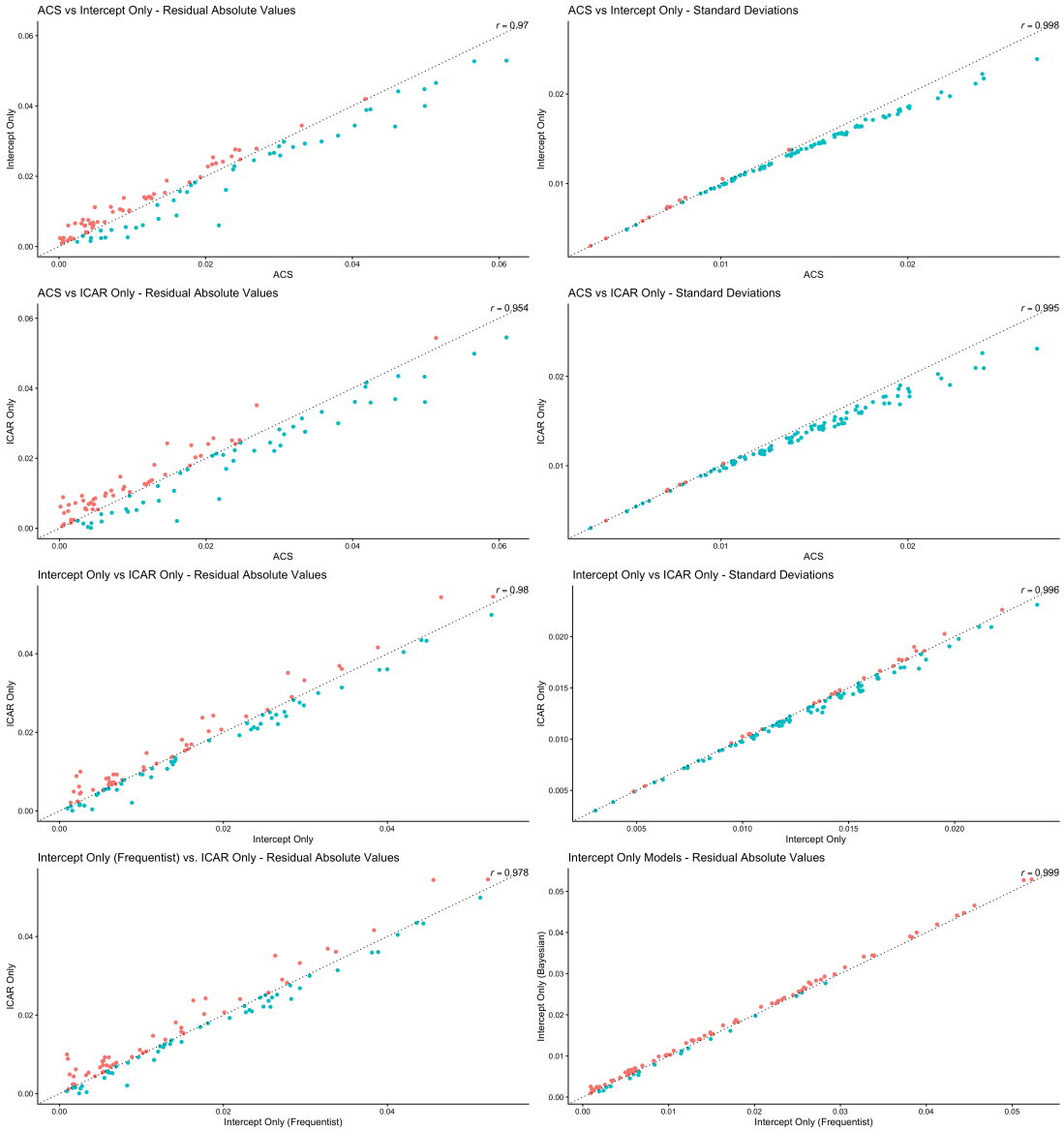


Figure 10: Plots comparing the residuals and standard deviations of vacancy rate estimates across models. Colors represent a points position relative to the reference line.

When comparing the two Fay-Herriot models directly, visual examination shows that the standard deviations are almost all lower for the ICAR model. It is worth noting that they stick extremely close to the trend line, so the difference isn't enormous. Similar behavior is seen for the residuals. It's a tossup for all of the counties, with more favoring the ICAR model. However, the difference is fairly slim and could very well just be due to random chance.

The bottom row of plots compares the Bayesian models to the Frequentist Intercept Only model. The bottom left plot with the ICAR model is, visually, nearly identical to the plot above it which examines the Bayesian Intercept Only model. The bottom right plot of the grid compares the two Intercept Only models, the Bayesian and Frequentist approaches. Almost all of the counties fall above the reference line, indicating that the Bayesian approach has slightly higher residuals than the frequentist approach across the board. They all fall very

close to the reference line however. This provides some light evidence in the direction of behavioral equivalence between Intercept Only models.

4.3.2. Interval Coverage Outcomes

When digging into improvements to residuals and standard deviations, it is always crucial to examine if the changes are meaningful and beneficial. To accomplish this, the interval coverage metric shown in Table 10 is examined more closely. Figure 11 shows counties that are gained and lost when moving to one of the two SAE models. Whether a county is captured or not is based on if the 95% prediction interval for a county contains the true value from the census.

Earlier in the report it was mentioned that the increased value of interval coverage relative to the ACS implies a gain with no tradeoff. This plot shows the tradeoff that summary metric fails to capture. The lefthand plot shows the outcomes resulting from the ICAR model prediction intervals. Of the 99 counties, 5 of them have true values that fall within the ICAR intervals that did not fall in the ACS intervals. However, 3 counties show the opposite result. These are counties captured by the ACS prediction intervals that fall outside the ICAR intervals. This results in a net gain of 2 counties.

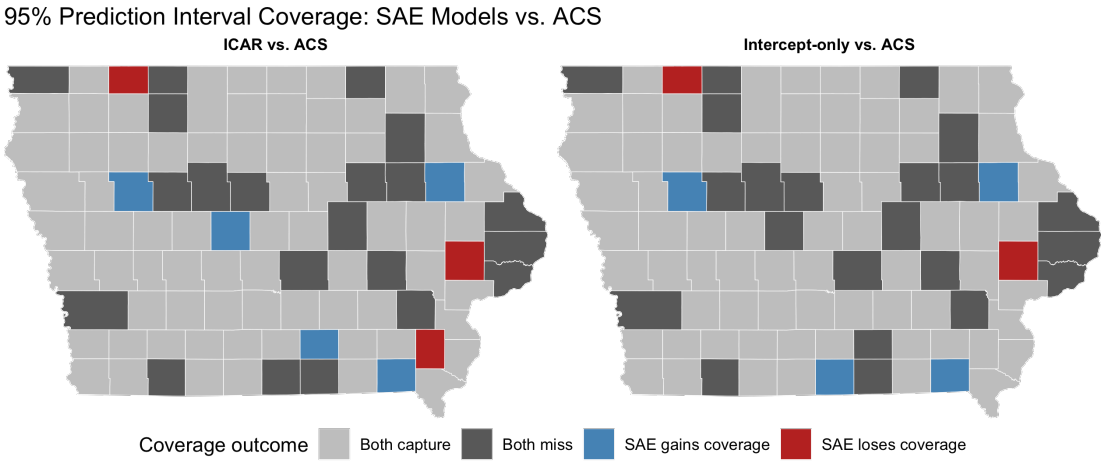


Figure 11: Plots showing the consequences of SAE estimates and intervals. Blue squares represent counties where the SAE intervals capture the true value that the ACS intervals failed to. Red is the opposite, showing true values that the ACS captured that the SAE interval failed to.

A particular limitation of the interval coverage metric appears when examining the Intercept only plot on the right. It shows the same net gain, 2 more counties gained than lost. However, that model captures 4 counties and loses 2. The interval coverage metric implies that these two models are equivalent, though they reach different conclusions across counties.

Both models are consistent across the vast majority of counties. They both gain Deleware, Sac, and Van Buren counties. They both lose Cedar and Dickinson counties. There are 4 counties where outcomes differ. Of those, the ICAR model gains Boone and Monroe county

but loses Henry county. The intercept only model gains Wayne county and does not lose Henry county.

5. Conclusion

Bibliography

[1] U.S. Census Bureau, "About the Decennial Census of Population and Housing." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/decennial-census/about.html

[2] U.S. Government Accountability Office, "2020 Census: Coverage Errors and Challenges Inform 2030 Plans." Accessed: Oct. 18, 2026. [Online]. Available: https://www.gao.gov/products/gao-25-107160

[3] U.S. Census Bureau, "About the American Community Survey." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about/acs-and-census.html

[4] U.S. Census Bureau, "The Importance of the American Community Survey and the Decennial Census." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about.html

[5] U.S. Census Bureau, "Top Questions About the Suvey." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about/top-questions-about-the-survey.html

[6] U.S. Census Bureau, "Using 1-Year or 5-Year American Community Survey Data." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/guidance/estimates.html

[7] USDA National Agricultural Statistics Service, "Cropland Data Layer." Accessed: Oct. 18, 2026. [Online]. Available: https://nassgeodata.gmu.edu/CropScape/

[8] U.S. Census Bureau, "Instructions for Applying Statistical Testing to American Community Survey Data." Accessed: Oct. 29, 2026. [Online]. Available: https://www2.census.gov/programs-surveys/acs/tech_docs/statistical_testing/2021_Instructions_for_Stat_Testing_ACS.pdf

[9] S. Cucu, A. Muhammad, S. M. Rijalus, C. A. Putri, and G. P. D. Sohibien, "Improving Precision in Small Area Proportion Estimation Using Logit Transformation: A Case of Internet Utilization in Papua's Regencies, Indonesia (2021)," *International Journal of Advances in Data and Information Systems*, vol. 7, no. 1, pp. 12–23, 2026.

[10] A. C. Staff, Accessed: Oct. 29, 2026. [Online]. Available: https://www.census.gov/library/stories/state-by-state/iowa.html

[11] Skforecast, Accessed: Nov. 06, 2026. [Online]. Available: https://skforecast.org/0.18.0/faq/probabilistic-forecasting-crps-score.html